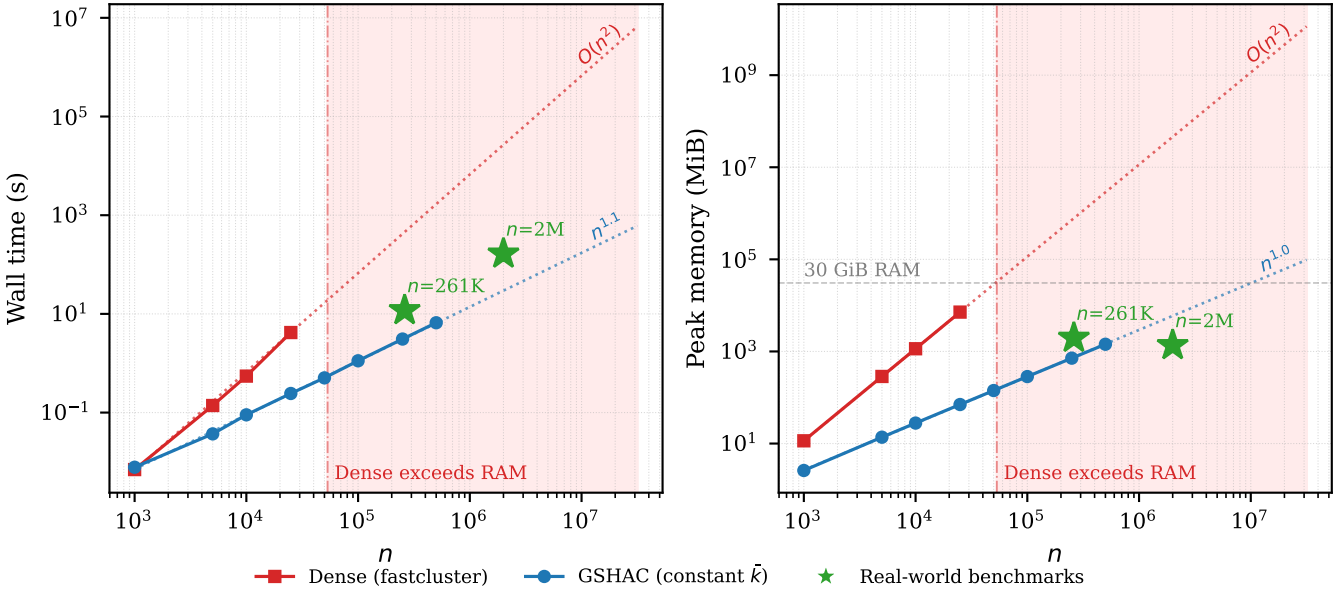
Wall time (s)
10^{7}
10^{5}
10^{3}
10^{1}
10^{-1}
$O(n^2)$
$n^{1.1}$
n=2M
n=261K
Dense exceeds RAM
10^{3}
10^{4}
10^{5}
10^{6}
10^{7}
n
Peak memory (MiB)
10^{9}
10^{7}
10^{5}
10^{3}
10^{1}
30 GiB RAM
$O(n^2)$
$n^{1.0}$
n=261K
n=2M
Dense exceeds RAM
10^{3}
10^{4}
10^{5}
10^{6}
10^{7}
n
Dense (fastcluster)
GSHAC (constant $\bar{k}$)
Real-world benchmarks